MILITARY SPECIFICATION

HEAT TRANSFER FLUID, FLUOROCHEMICAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers inert, fluorochemical, liquid heat transfer agents.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290	Packaging, Packing and Marking of Petroleum Products

(Copies of specifications and standards required by Suppliers in Connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Publications

D86	Boiling Range
D97	Pour Point
D445	Viscosity
D877	Dielectric Strength
D1298	Specific Gravity

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(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material The heat transfer fluid shall be a clear, colorless, odorless, non-flammable, inert fluorochemical liquid.

3.2 Physical properties The physical properties of the heat transfer fluid shall conform to the requirements of Table I:

TABLE I
Physical Properties

Property	Limits	Test Paragraph
Boiling range 194°F(90°C) to 225°F(107°C)	90 percent (minimum)	4.4.1
Viscosity		4.4.1
77°F(25°C)	0.50 centistokes(min.)	
-40°F(-40°C)	5.0 centistokes (max.)	
Pour point	Less than -80°F(-62°C)	4.4.1
Specific gravity 25°C/25°C	1.72 ± 0.08	4.4.1
Hydrolytic Stability	10 micrograms per milliter fluoride (max.)	4.4.2

3.3 Electrical properties The electrical properties shall conform to the requirements of Table II:

TABLE II
Electrical Properties

Property	Limit	Test Paragraph
Dielectric Strength at 77°F (25°C)	340 V/mil(min)	4.4.1
Electrical Resistance	2×10^5 megohms(min)	4.4.3

3.4 Workmanship The heat transfer fluid shall be uniform and free from dirt and other foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for quality conformance inspection and tests shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Sampling for examination of end item. A random sample of filled containers and a sample of shipping containers fully prepared for delivery shall be selected from each lot of heat transfer fluid in accordance with MIL-STD-105, inspection level I.

4.2.2 Quality conformance inspection.

4.2.2.1 Examination of end item. The end item shall be visually examined for defects in appearance and workmanship (see paragraphs 3.1 and 3.4). The lot size, for determining the sample size, shall be expressed in units of filled containers. The sample unit shall be one filled unit container. The inspection level shall be level I and Acceptable Quality Level (AQL) of 2.5 expressed in defects per hundred units.

4.2.2.2 Examination of filled containers for net contents. The sample unit for this examination shall be one filled unit container. The inspection level shall be S-2. The average net contents per container for all sample units examined shall be not less than quantity specified.

4.2.2.3 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and markings comply with the requirements of Section 5 of this specification. Table III indicates the major criteria. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. The inspection level for determining the sample size shall be S-2 and the acceptable quality level (AQL) shall be 2.5 expressed in defects per 100 units.

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TABLE III Major Inspection Criteria

EXAMINE	DEFECT
Packaging (as applicable)	Container not as specified. Container damaged, leaking, or otherwise defective. Closure not as specified.
Packing (as applicable)	Not level specified, not in accordance with contract requirements. Arrangement of units per shipping container not as specified. Container not as specified, closures not accomplished by specified methods or materials. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components, such as incomplete closures of case liners, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count (as applicable)	Less than specified or indicated quantity of primary containers per shipping container.
Markings	Omitted, incorrect, illegible, incomplete or not in accordance with contract requirements. Precautionary markings missing or not as specified.

4.3 Testing of the end item. The end item shall be tested in accordance with the tests specified in 4.4 for conformance to the specified requirements. The sample unit for test shall be one pint of material. The sample size shall be S-1 with no less than three sample units randomly selected from the lot. There shall be no failure to meet the specified requirements.

4.4 Test Methods

4.4.1 The tests listed in Table IV shall be conducted in accordance with the applicable ASTM Test Methods.

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TABLE IV
Test Methods

Test	ASTM Method
Dielectric strength	D877
Pour point	D97
Specific Gravity	D1298
Viscosity	D445
Boiling Range	D86

4.4.2 Hydrolytic Stability

4.4.2.1 Reagents - Dissolve 0.5 gram of zirconium oxychloride ($ZrOCl_2 \cdot 8H_2O$) in distilled water in a 100 ml volumetric flask, dilute to mark and mix well. Dissolve 0.1 gram of alizarin sodium monosulfate in 100 ml. distilled water and store in an amber colored bottle in a cool place. To 25 ml of the zirconium oxychloride solution add dropwise and with agitation, 25 ml of the alizarin monosulfate solution. If a turbidity persists allow the mixture to stand until it settles, usually overnight. Then add 100 ml. of distilled water and mix well. Store the diluted solution in an amber colored bottle in a cool place. Prepare a fresh diluted solution whenever needed, discarding the solution when it is more than 72 hours old.

4.4.2.2 Fluoride standard Dissolve 0.2210 g of CP sodium fluoride in distilled water in a 100 ml volumetric flask, dilute to volume and mix well. This solution contains 0.1 percent or 1000 ppm of fluoride. When 5 ml of this solution is diluted to 100 ml. in a volumetric flask and mixed well the diluted solution contains 50 ppm fluoride.

4.4.2.3 Procedure

4.4.2.3.1 Hydrolysis Twenty five ml of a 1 percent $NaHCO_3$ solution and 25 ml of the sample shall be placed in a 250 ml flask fitted with a reflux condenser. The refluxing shall last four hours using boiling chips and a heating mantle and minimum heat to maintain the refluxing. The liquid in the flask shall be cooled and the aqueous (upper) layer separated. The fluorochemical (bottom) layer shall be washed with three 10 ml portions of distilled water. Combine the aqueous layer and washings, and dilute to 120 ml with distilled water.

4.4.2.3.2 Fluoride determination Fifty ml of the aqueous solution from 4.4.2.3.1 shall be placed in a 50 ml Nessler tube. To five additional Nessler tubes shall be added 0.0, 4.0, 5.0, 6.0, 10.0 and 12.0 ml. of the standard sodium fluoride solution and then diluted to 50 ml with distilled water. (These standards correspond

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to 0,4,5,6,10 and 12 micrograms per milliliter of fluorine respectively). The contents of each Nessler tube shall be mixed with a plunger type stirrer, such as a glass rod flattened at one end. Then 5 ml of 5N hydrochloric acid shall be added to each of the six Nessler tubes containing the sample and five standards. The solutions shall be mixed well. Then one ml of the zicronium alizarin indicator shall be added to each Nessler tube and mixed again. The standards and sample shall be allowed to stand at room temperature for about 18 hours, mixed again and compared to determine fluoride content of sample.

4.4.3 Dielectric strength The dielectric strength shall be determined in accordance with ASTM Method D-877, alternative method.

4.4.3.1 Electrical resistance The electrical resistance shall be determined by placing a variable capacitor of 11 to 100 mmf (similar to equivalent to J.D. Cardwell 5) in a 400 ml beaker. The plates shall be covered with the test sample. The plates shall be completely meshed, and care should be taken that there are no entrapped bubbles. The beaker shall be placed in a safety test box and the capacitor leads shall be connected to the "test" and "Hi" terminals on a megohm meter (similar or equivalent to Industrial Instrument Model L-7) that has been set on X10K and allowed to warm up for 30 minutes. The switch shall be set to volts and the voltage adjusted to 500 volts. The switch shall be returned to megohms and the galvanometer needle adjusted to infinity. The calibration switch shall be depressed and the calibration adjusted to "1". The last two steps shall be repeated until adjustment is no longer necessary. A reading shall be taken after two minutes. The switch shall be depressed and the meter read. The reading shall be multiplied by 10,000. If the reading is greater than 500 times 10^4 it shall be reported as greater than 5×10^6 megohms. The switch shall be released and the multiplier set as "off".

CAUTION

A capacitor can store a large electrical charge. Be sure the shorting mechanism on the safety box is operating properly before attempting to remove the leads. Let the capacitor stand 30 minutes in reagent grade acetone; then place the capacitor in an oven at 300°F(149°C) for one hour. Cool in a desiccator.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking Packaging, packing, and marking of the heat transfer fluid shall be in accordance with MIL-STD-290. The type and size of the unit containers shall be as specified by the procuring activity (see 6.2).

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6. NOTES

6.1 Intended use. The fluorocarbon fluid covered by this specification is intended for use in closed vapor tight systems as a heat transfer medium having superior electrical insulating and dielectric properties. The temperature range as a liquid coolant is from -112°F(-80°C) to 221°F (105°C) and as an evaporative coolant up to (752°F) 400°C.

6.2 Ordering data.

(a) Title and number of this specification
(b) Quantity desired
(c) Type and size of containers
(d) Applicable level of packaging and packing

6.3 Products known to meet the requirements of this specification are "FC-77" and "FC-75" made by Minnesota Mining and Manufacturing Company St. Paul, Minnesota 55119 and Freon E-2 made by E.I. Dupont de Nemours & Co., Wilmington, Delaware 19898.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification 14755 dated 1 March 1967.

PREPARING ACTIVITY
Navy - AS

Custodian
Army - MU
Navy - AS
AF - 68

Proj. No. 6850-0372

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-H-81829 HEAT TRANSFER FLUID, FLUOROCHEMICAL

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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